

Susceptibility Class	Recommended Color	Description / Interpretation			
Very Low	Dark Green	Stable areas with flat terrain, dense forest cover, and low rainfall.			
Low	Light Green	Areas with gentle slopes and minimal anthropogenic disturbance.			
Moderate	Yellow	Transitional zones where susceptibility is balanced by environmental factors.			
High	Orange	Highly susceptible areas with steep slopes, high rainfall, or sparse vegetation.			
Very High	Red	Critical zones near road-cuts, active streams, or extremely steep barren slopes.			

Technical and Geomorphological Justification of Weight Distribution in Landslide Susceptibility Mapping (LSM)

1. Introduction

This report presents the technical and geomorphological justification for the weight distribution adopted in the Landslide Susceptibility Mapping (LSM) model. The weights were derived using the **Analytic Hierarchy Process (AHP)**, a multi-criteria decision-making technique that systematically ranks influencing factors based on their relative importance.

The objective of this weighting scheme is to ensure that each parameter contributes to the susceptibility model according to its scientifically established role in slope instability.

2. Mathematical Foundation of the Weighting Scheme

2.1 Analytic Hierarchy Process (AHP)

The weighting values were computed from a **12 × 12 pairwise comparison matrix** using the eigenvector method proposed by Thomas L. Saaty (1980).

The principal eigenvector of the comparison matrix was extracted to determine the relative weights (w) of each conditioning and triggering factor.

Each percentage value represents the normalized priority of a factor in contributing to landslide occurrence.

2.2 Consistency Verification

To ensure logical reliability of expert judgments, the **Consistency Ratio (CR)** was calculated.

- Calculated CR = **0.079**
- Acceptable threshold = **CR < 0.10**

Since the CR value is below 0.10, the comparison matrix is considered **scientifically consistent**, indicating that the assigned priorities are logically sound and free from major inconsistency.

3. Scientific Justification of Weight Distribution

The assigned weights are grounded in geomorphological theory and landslide mechanics. Each parameter is categorized as either:

- **Primary Trigger**
 - **Primary Conditioning Factor**
 - **Secondary Modifier**
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4. High-Weight Factors (Primary Controls)

4.1 Slope (21.1%)

Slope received the highest weight (21.1%), reflecting its fundamental role in slope instability.

According to David J. Varnes (1984), slope angle directly governs the balance between:

- **Shear stress (driving force due to gravity)**
- **Shear strength (resisting force from soil friction and cohesion)**

As slope angle increases:

- Gravitational force increases
- Internal resistance becomes insufficient
- Probability of failure significantly rises

Thus, slope is considered the **primary geomorphological driver** of landslides.

4.2 Rainfall (16.8%)

Rainfall received the second-highest weight (16.8%) because it acts as the **dynamic environmental trigger**.

Research by Norman Caine (1980) demonstrated that rainfall intensity–duration thresholds are the most common cause of shallow landslides and debris flows.

Rainfall contributes to instability by:

- Increasing pore-water pressure

- Reducing effective stress
- Decreasing shear strength
- Promoting rapid slope failure

Unlike slope (a static factor), rainfall is a **temporal trigger**, making it critically important in susceptibility modeling.

5. Mid-Tier Factors (Significant Conditioning Controls)

5.1 Land Use / Land Cover (LULC) – 12.6%

LULC influences surface stability and anchoring conditions.

- Forested areas enhance root cohesion.
- Barren land lacks mechanical reinforcement.
- Built-up areas increase runoff and modify drainage patterns.

Vegetation plays a stabilizing role; however, its influence depends on slope and rainfall conditions. Therefore, LULC was assigned a mid-level weight.

5.2 Topographic Wetness Index (TWI) – 9.8%

TWI identifies zones of potential soil saturation by integrating:

- Slope gradient
- Flow accumulation

High TWI values indicate areas prone to prolonged moisture retention, which:

- Elevates pore-water pressure
- Weakens soil structure
- Promotes internal slope failure

TWI is more informative than simple drainage metrics because it quantitatively models hydrological convergence.

6. Toe-Erosion and Anthropogenic Influence

6.1 Distance to Streams (7.9%)

Streams contribute to slope instability through **toe erosion** and undercutting. Removal of basal support increases susceptibility to mass movement.

6.2 Distance to Roads (6.3%)

Road construction often involves slope cutting and excavation. These activities:

- Remove structural support
- Alter natural drainage
- Create artificial steep slopes

According to A. K. Pachauri and M. Pant (1992), anthropogenic slope modification significantly increases landslide probability.

Both parameters were assigned higher weights than spectral indices due to their direct mechanical impact on slope stability.

7. Low-Weight Factors (Secondary Modifiers)

7.1 NDVI (3.6%)

NDVI reflects vegetation health and density. However:

- Its effect overlaps with LULC classification.
- It primarily refines surface conditions rather than acting as a primary driver.

Thus, NDVI serves as a fine-tuning parameter rather than a dominant control factor.

7.2 Drainage Density (3.0%)

Drainage density represents runoff concentration potential. However:

- High drainage density alone does not trigger landslides.
- It becomes significant only in conjunction with steep slopes and high rainfall.

Therefore, it was categorized as a secondary conditioning factor.

8. Comparative Scientific Logic of Weight Hierarchy

Parameter A	>	Parameter B	Scientific Justification
Slope	>	Elevation	Slope directly controls shear stress; elevation only influences climatic regime.
Rainfall	>	NDVI	Rainfall triggers failure; NDVI reflects vegetation condition only.
Roads	>	Aspect	Road cuts cause immediate mechanical instability; aspect influences microclimate over long time scales.
TWI	>	Curvature	TWI integrates hydrology and slope; curvature only describes surface shape.

9. Conclusion

The weight distribution derived through AHP reflects both **mathematical rigor** and **geomorphological reasoning**. The acceptable Consistency Ratio confirms logical coherence in expert judgment.

The hierarchy clearly distinguishes:

- **Primary drivers:** Slope and Rainfall
- **Major conditioning factors:** LULC and TWI
- **Anthropogenic and hydrological modifiers:** Roads and Streams
- **Secondary refinements:** NDVI and Drainage Density

This scientifically justified weighting framework strengthens the reliability of the Landslide Susceptibility Model and ensures that the generated susceptibility zones accurately represent real-world slope instability mechanisms.

10. References

- Thomas L. Saaty (1980). *The Analytic Hierarchy Process: Planning, Priority Setting, Resource Allocation*. McGraw-Hill.
- David J. Varnes (1984). *Landslide Hazard Zonation: A Review of Principles and Practice*. UNESCO, Paris.
- Norman Caine (1980). *The Rainfall Intensity-Duration Control of Shallow Landslides and Debris Flows*. Geografiska Annaler.

- A. K. Pachauri & M. Pant (1992). *Landslide Hazard Mapping Based on Geological Attributes*. Engineering Geology.

1. Topographic & Morphological Factors (DEM Derived)

These parameters define the physical structure and “stage” of the terrain.

1.1 Slope (21.1% Weight)

- **Scientific Logic:** Slope is the primary driver of landslides; as the angle increases, the tangential component of gravity (shear stress) increases while the normal component (friction) decreases.
- **Classification:**
 - **0° – 5.7°:** Rank **1** (Stable plains).
 - **34° – 58°:** Rank **5** (Critically steep; gravity easily overcomes soil cohesion).

Old Values (Range)	New Value (Rank)	Susceptibility
0 - 5.7	1	Very Low
5.8 - 14	2	Low
15 - 23	3	Moderate
24 - 33	4	High
34 - 58	5	Very High

1.2 Elevation / DEM (5.2% Weight)

- **Scientific Logic:** Elevation influences local climate, weathering rates, and moisture. Higher altitudes often feature steeper, less stable slopes and thinner soil cover.
- **Classification:**
 - **180m – 280m:** Rank **1**.
 - **520m – 750m:** Rank **5**.

1.3 Topographic Wetness Index (TWI) (9.8% Weight)

- **Scientific Logic:** TWI quantifies topographic control on hydrological processes, indicating areas where water is likely to accumulate and saturate the soil, increasing pore-water pressure.
- **Classification:**
 - **–8.7 to –6:** Rank **1**.
 - **2.3 to 8.7:** Rank **5**.

Old Values (Range)	New Value (Rank)	Susceptibility
-8.7 - -6	1	Very Low
-5.9 - -4.2	2	Low
-4.1 - -1.4	3	Moderate

Old Values (Range)	New Value (Rank)	Susceptibility
-1.3 - 2.2	4	High
2.3 - 8.7	5	Very High

2. Geo-Environmental & Vegetation Factors

These factors determine the “anchoring” strength and physical composition of the surface.

2.1 NDVI (3.6% Weight)

- **Scientific Logic:** Vegetation anchors soil through root systems and reduces saturation via transpiration. Low NDVI (barren land) lacks this protection.
- **Classification (Inverse):**
 - **0.26 – 0.35:** Rank **1** (Dense vegetation).
 - **0.024 – 0.13:** Rank **5** (Barren/exposed soil).

Old Values (Range)	New Value (Rank)	Susceptibility
0.26 - 0.35	1	Very Low
0.23 - 0.25	2	Low
0.19 - 0.22	3	Moderate
0.14 - 0.18	4	High
0.024 - 0.13	5	Very High

2.2 Land Use/Land Cover (12.6% Weight)

- **Scientific Logic:** Forested areas provide root cohesion; barren and built-up areas increase runoff and lack mechanical stabilization.

Category	New Value (Rank)	Susceptibility
Water / Forest	1	Very Low
Crop	2	Low
Built-up	4	High
Barren Land	5	Very High

2.3 Soil Texture (4.4% Weight)

- **Scientific Logic:** Soil texture determines drainage and shear strength. Clay-heavy soils (Clay Loam) expand when wet, increasing instability.

[Image of the USDA soil texture triangle]

Category	New Value (Rank)	Susceptibility
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Category	New Value (Rank)	Susceptibility
Silty Clay Loam	3	Moderate
Sandy Clay Loam	4	High
Clay Loam	5	Very High

3. Hydrological & Triggering Factors

These represent the active forces that initiate mass movement.

3.1 Precipitation / Rainfall (16.8% Weight)

- **Scientific Logic:** Rainfall is the primary triggering factor. High accumulation saturates the failure plane and increases the weight of the slope.
- **Classification:**
 - **2037 mm:** Rank 1.
 - **2098 – 2163 mm:** Rank 5.

3.2 Distance to Stream (7.9% Weight)

- **Scientific Logic:** Proximity to streams is dangerous due to fluvial “toe-erosion,” which removes the natural support at the base of a slope.
- **Classification:**
 - **2.7 – 4.4 km:** Rank 1.
 - **0 – 0.42 km:** Rank 5 (Active erosion zone).

Old Values (Range)	New Value (Rank)	Susceptibility
2.7 - 4.4	1	Very Low
1.7 - 2.6	2	Low
0.93 - 1.6	3	Moderate
0.43 - 0.92	4	High
0 - 0.42	5	Very High

3.3 Drainage Density (3.0% Weight)

- **Scientific Logic:** High drainage density reflects highly dissected terrain with high surface runoff potential.

Old Values (Range)	New Value (Rank)	Susceptibility
7,000 - 8,500	1	Very Low
5,300 - 6,900	2	Low
3,700 - 5,200	3	Moderate
2,000 - 3,600	4	High

Old Values (Range)	New Value (Rank)	Susceptibility
280 - 1,900	5	Very High

4. Anthropogenic Factor (Infrastructure)

4.1 Distance to Road (6.3% Weight)

- **Scientific Logic:** Road construction involves cutting into the “toe” of mountainsides, creating unsupported vertical faces prone to slumping.
- **Classification:**
 - **1.3 – 1.9 km:** Rank 1.
 - **0 – 0.2 km:** Rank 5 (Maximum disturbance zone).

Old Values (Range)	New Value (Rank)	Susceptibility
1.3 - 1.9	1	Very Low
0.77 - 1.2	2	Low
0.47 - 0.76	3	Moderate
0.21 - 0.46	4	High
0 - 0.2	5	Very High

Summary Checklist for Final Map Generation:

- All 12 layers reclassified to 1–5 scale.
- **Raster Calculator** formula applied with AHP weights.
- Result categorized into 5 zones using **Natural Breaks (Jenks)**.